

Assignment 03

Vite Module Bundler + DSA Practice

Build a small Vite app and practice Search & Sort algorithms

Advanced JavaScript Course

No solutions required - submit requirements only

Assignment Overview

Main Goal

- Use Vite as the module bundler / build tool, not Webpack.
- Practice ES Modules by splitting code into multiple files using import/export.
- Use real Vite features: dev server, production build, env variables, static assets, CSS imports, and aliases.
- Apply the DSA topics covered so far: searching and sorting algorithms.

Part 1 - Create a Vite App

- Create a new Vanilla JavaScript project using Vite.
- The project should run locally using the Vite dev server.
- The final project must include a production build using Vite build.
- Do not use Webpack in this assignment.

Expected starter commands:

```
npm create vite@latest
npm install
npm run dev
npm run build
npm run preview
```

Part 2 - Required Folder Structure

Students can add more files, but the project should at least include this structure:

```
src/
  main.js
  data/
    students.js
  utils/
    search.js
    sort.js
  components/
    renderList.js
  styles/
    main.css
public/
  vite.svg
.env
```

Part 3 - Use Vite Features

Required Vite Features

- Run the app using Vite dev server with `npm run dev`.
- Import CSS directly inside JavaScript using `import './styles/main.css'`.
- Import and render at least one asset/image from `src` or `public`.
- Use an environment variable that starts with `VITE_`, for example `VITE_APP_TITLE`.
- Create a production build using `npm run build`.
- Test the production build locally using `npm run preview`.
- Bonus: configure an alias like `@utils` in `vite.config.js` and use it in imports.

Part 4 - App Feature Requirements

Build a small UI that displays a list of items. The list can be students, products, courses, or tasks.

UI Requirements

- Display the original data list on the page.
- Add a search input and show whether the searched item exists or not.
- Add buttons to sort the data using different algorithms.
- Show the result after sorting without breaking the original data.
- Show a small label for the current algorithm used.
- Use separate modules for data, search, sort, and rendering.

Important rule:

Do not write all code in main.js. The goal is to practice modules and bundling, so the logic must be split across files.

Part 5 - DSA Tasks

Implement the following functions in JavaScript. You can connect them to the UI or keep them in separate utility files and call them from the app.

Algorithm	Requirement	Expected Return
Linear Search	Search for a value inside an array one by one.	Index or -1
Binary Search	Search inside a sorted array using low, high, and mid.	Index or -1
Bubble Sort	Sort an array using repeated adjacent swaps.	Sorted array
Selection Sort	Sort by selecting the minimum item each round.	Sorted array
Insertion Sort	Sort by inserting each item in its correct position.	Sorted array

Example function names:

```
linearSearch(array, target)
binarySearch(sortedArray, target)
bubbleSort(array)
selectionSort(array)
insertionSort(array)
```

Part 6 - README Requirements

Create a README.md file and explain the following points in your own words:

README Checklist

- What is Vite?
- What problem does a module bundler / build tool solve?
- What happens when we run npm run dev?
- What happens when we run npm run build?
- What is the difference between development build and production build?
- What is the difference between Linear Search and Binary Search?
- Why does Binary Search require a sorted array?
- Write the Big-O for Linear Search, Binary Search, Bubble Sort, Selection Sort, and Insertion Sort.
- Mention which Vite features you used in your project.

Bonus Tasks

Optional Bonus

- Add vite.config.js and configure an alias such as @utils or @components.
- Add a small loading state while rendering data.
- Add a reset button to restore the original list.
- Show the time taken by each sorting algorithm using performance.now().
- Deploy the app using any static hosting provider.

Submission

Item	Required
Source code	Yes
Screenshot of running app	Yes
Screenshot of dist folder after build	Yes
README.md	Yes
Deployed link	Bonus

Note for students: The assignment is about understanding the flow, not copying code. Keep your files clean and commit your work step by step.